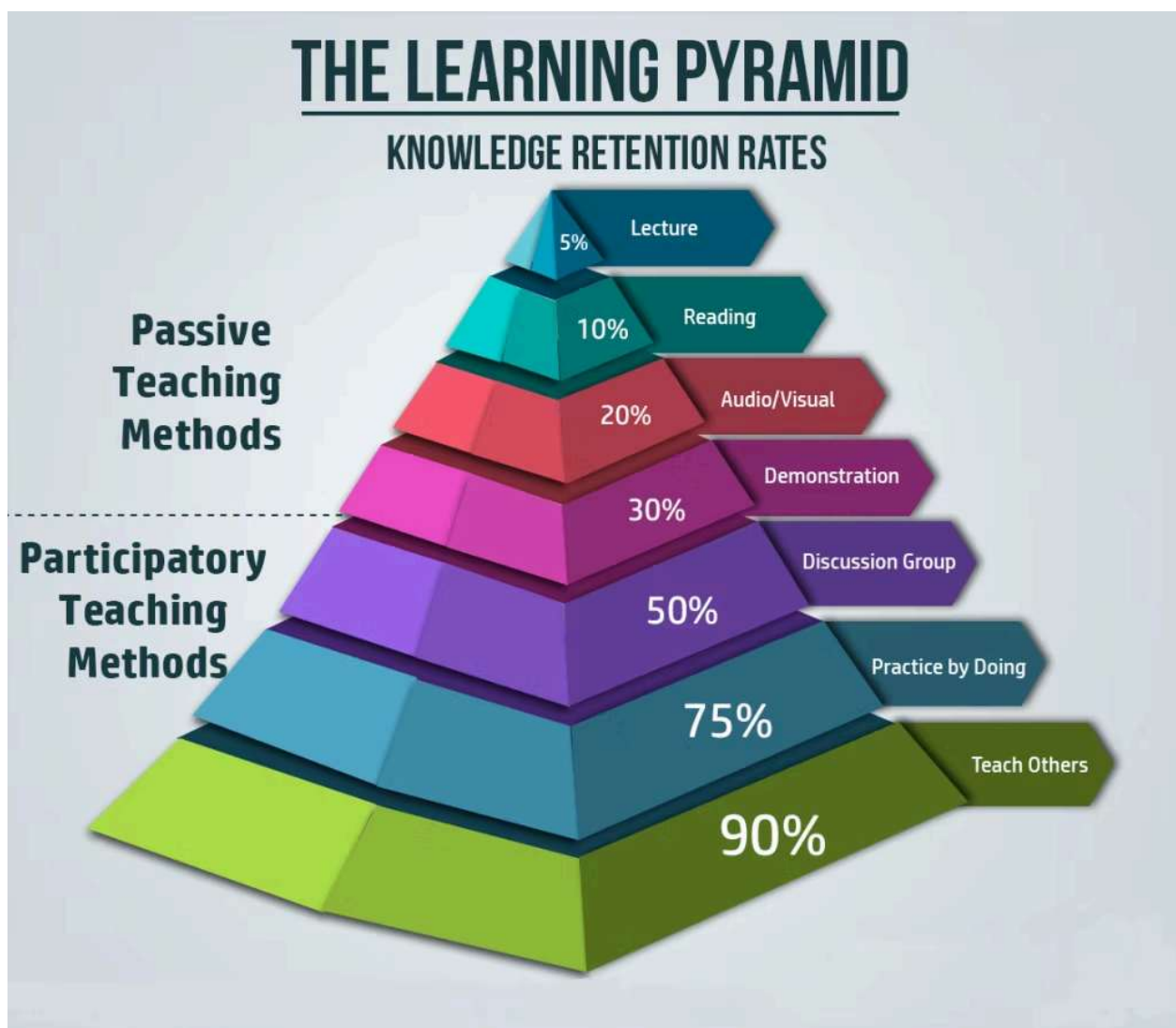




Pushing Your Teaching Down the Learning Pyramid

Updated: Oct 21, 2025

The Learning Pyramid model suggests that some methods of instruction are more effective than others and that varying instructional methods will lead to deeper learning and longer-term retention.



Source: Adapted from the National Training Laboratories Institute of Applied Behavioral Science Learning Pyramid

The Learning Pyramid developed by the National Training Laboratories Institute of Applied Behavioral Science illustrates that most students only remember about 5% of what they learn while listening to their teacher talk, but retain nearly 90% of what they learn through teaching others. This doesn't imply that we should abandon lectures altogether. Direct instruction still holds significant value. Instead, it suggests that it should be just one of many teaching methods used in the classroom. **There is a synergism between instructional methods, and the passive learning activities should set up the participatory learning activities.**

There is considerable academic debate regarding the origins of the Learning Pyramid and its neatly organized knowledge retention rates. The data from the National Training Laboratories Institute study done in the early 1960s to

support the specific knowledge retention percentages used in the model was lost. Some have suggested the model should be scrapped altogether. We disagree. While the Learning Pyramid is probably an oversimplified illustration, it generally tracks with our own learning experiences and observations in the classroom.

The scientific basis for the maximum allowable length of TED Talks - 18 minutes - is based upon the energy demands of authentically listening. As the brain takes in new information and is forced to process it, millions of neurons are firing at once, burning energy and leading to fatigue and exhaustion. After 18 minutes, the attention of your teenage students is only feigned. Fortunately, research also shows that a change in learning activity structure (ex: a shift from lecture to students talking or actively practicing newly-acquired skills) activates different neural pathways and more authentic learning can take place.

Abraham Lincoln's Gettysburg Address was about two minutes long. John Kennedy inspired a nation to look to the stars in 15 minutes. Dr. Martin Luther King Jr. took a bit longer to share his dream of racial equality—he did it in 17 minutes. We can explain a disciplinary core idea to our students in less than 18 minutes. Then we can lead our students in another learning activity where they apply that core idea to a real-world scenario. Guided student discussion, practice of newly-acquired skills, and hands-on investigations are the true hallmarks of an incredibly effective science educator, not another, albeit pretty, PowerPoint presentation.

The most effective teachers utilize a diverse set of passive and participatory instructional methods, with more time devoted to participatory methods. Case studies, practice applying newly learned skills, hands-on investigations, student presentations, and Process Oriented Guided Inquiry Learning (POGIL) activities are some of the most effective strategies to push your teaching down the learning pyramid and increase student learning, long-term retention, and enjoyment in your class.



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